SEC 6 · LEC 3 · ATTITUDE ACTUATORS

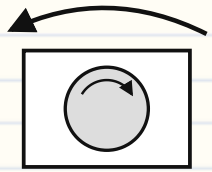
Spin a wheel, turn the ship

Left = your left notebook page. Right = your right page. Copy the drawing in pencil first, then the cues and notes. The summary box stays empty until the video and this PDF are closed.

LEFT PAGE · DRAW

S6 / L3 — Attitude actuators **date · pg _**

1 · INSIDE: wheels swap spin with the craft

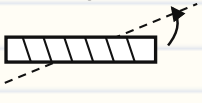


REACTION WHEEL
change the wheel's SPEED
→ the craft turns the other way



CONTROL MOMENT GYRO
speed stays the same
TILT its axis → a big twist

2 · OUTSIDE: push on something else

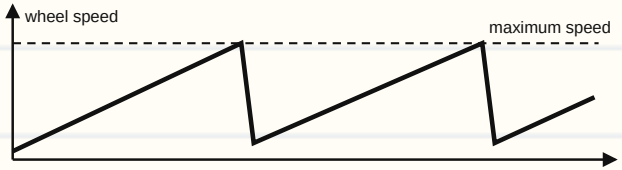


MAGNETIC TORQUER
a coil = an electromagnet
Earth's field twists it, like a compass
weak · no fuel · low orbit only



THRUSTERS
small jets, off-centre, in pairs
strong · but they use propellant

3 · WHEELS FILL UP (saturation)



ramp = soaking up a steady outside nudge · drop = dumping it

wheels and CMGs only SWAP spin with the craft
thrusters and magnetic torquers REMOVE it

✏️ **PENCIL KEY** · thin arrow = wheel or exhaust · thick arrow = the craft's turn · dashed arrow = tilting · dashed line = magnetic field or limit

⭐ **3 TAKEAWAYS** 3 lines

1. Spin a wheel one way and the spacecraft turns the other way.
2. Wheels and CMGs only swap spin, so they fill up (saturate).
3. Thrusters and magnetic torquers push on the outside and empty them.

🧠 **SUMMARY · video + PDF closed** 3 lines

1. _____
2. _____
3. _____

? RIGHT PAGE · CORNELL

CUES (questions)	NOTES (keywords)
The five actuators?	actuators twist the spacecraft: momentum wheels · reaction wheels · magnetic torquers · control moment gyros · thrusters
Why does spinning a wheel turn the spacecraft?	conservation of angular momentum: spin a wheel one way → the craft turns the other (helicopter + tail rotor)
Momentum wheel vs reaction wheel?	momentum wheel: always spinning, 5,000+ rpm · change its speed to turn the craft · fast, so the wheel can be light
What is saturation? What fixes it?	saturation: a wheel at top speed can do no more → thrusters or other help
Why 3 or 4 wheels?	DC motor → flywheel → spacecraft rotation · too fast → the wheel can break
CMG: what changes? Why use one?	reaction wheel: same idea, starts at rest · change the SPEED · 3–4 wheels for 3 axes
Magnetic torquer: how? Good and bad?	CMG: wheel at constant speed · change the DIRECTION of its axis → large torque · high slew rate · the ISS uses them
Thrusters: how do they make a twist?	magnetic torquer: coil (solenoid) makes a magnet that the field of the Earth twists · weak · no moving parts, reliable, cheap
What is slew rate?	thrusters: small cold-gas or hot-gas jets placed off-centre · torque = force × arm · use propellant
? which ones can empty a wheel	

SEC 6 · LEC 3 · TEACH

Turning with nothing to push on

Simple story first, then the official names, then how it actually works with numbers you can check.

🧒 Like you're 5

Sit on a swivel chair with your feet up and swing your arms hard to the left. The chair turns to the right. You pushed on nothing; you and your arms just traded spin. That is a **reaction wheel**: a motor spins a heavy wheel one way and the whole spacecraft slowly turns the other way. Stop the wheel and the spacecraft stops, pointing somewhere new. No fuel at all.

The catch: sunlight and thin air keep nudging the spacecraft the same way, day after day. To hold still, the wheel must keep spinning faster to soak that up, until it hits top speed. It is full: **saturated**. To empty it you must push on something *outside*: a puff from a small jet (**thruster**), or an electromagnet that leans on the Earth's magnetic field like a compass needle (**magnetic torquer**). For very big spacecraft there is a stronger trick. Hold a spinning bicycle wheel by its axle and try to tilt it: it twists hard in your hands. A **control moment gyro** is that wheel, tilted on purpose. Four of them steer the Space Station.

🏷 The official names (name first, then what it means)

- **Actuator**: any device that applies a torque. **Momentum-exchange devices**: the ones that work by trading spin (wheels and CMGs).
- **Reaction wheel** (RW; the slide's "reactor wheel" is a typo): a wheel that normally sits near zero speed and is spun either way.
- **Momentum wheel** (bias momentum wheel): a wheel kept spinning fast all the time. Its spin also stiffens the spacecraft against tipping: **gyroscopic stiffness**. **Flywheel**: the heavy rotor itself.
- **Angular momentum** (h or L): moment of inertia × spin rate, in newton-metre-seconds. **Conservation of angular momentum**.
- **Saturation**: a wheel at its top speed. **Momentum dumping** (desaturation, momentum unloading): emptying it with an outside torque.
- **Control moment gyroscope, CMG**: a wheel at constant speed on a motor-driven **gimbal**. Output torque = angular momentum × **gimbal rate**.
- **Slew**: a deliberate turn to a new target. **Slew rate**: how fast it can be done, in degrees per second.
- **Magnetorquer** (magnetic torquer, torque rod, torque coil): a coil that makes a **magnetic dipole moment**; the planet's field twists it.
- **Reaction control system, RCS**: the set of small attitude thrusters. **Cold-gas thruster** (just releases stored gas) and **monopropellant thruster** (hot gas). A **couple**: two equal thrusters on opposite sides, giving a pure twist. **Moment arm**: distance from the centre of mass.
- **Internal torque** (wheels, CMGs) versus **external torque** (thrusters, magnetorquers, and the disturbances).

🛠 How it actually works

Step	The mechanism, with numbers
1 · The swap	Wheel spin + craft spin = constant. Wheel inertia 0.01, craft 100 kg·m²: a ratio of 10,000. Spin the wheel up to 1,000 rpm and the craft turns at 0.1 rpm (0.6° per second) the other way. Brake the wheel and the craft stops.
2 · Three axes	One wheel per axis, plus a fourth mounted at a slant as a spare. The Kepler telescope lost two of its four in 2013 and could no longer hold its target.
3 · Filling up	A steady outside twist of 0.00001 N·m for a day (86,400 s) adds 0.86 N·m·s. A wheel that holds 4 N·m·s is full in under **5 days**.
4 · Dumping	Switch on a magnetorquer or fire a thruster to twist the craft one way while the wheel slows down to cancel it. The extra spin leaves the spacecraft for good.
5 · CMG	The Space Station's: a 98 kg wheel at 6,600 rpm stores 4,760 N·m·s. Tilt its axis at 3.1° per second (0.054 rad/s) and out comes 4,760 × 0.054 ≈ **258 N·m**. A reaction wheel gives about 0.1 N·m.
6 · Magnetorquer	Torque = coil strength × field: 10 A·m² × 0.00003 tesla = **0.0003 N·m**. Tiny but free. It cannot twist about the field line itself, and it needs a strong field: low orbit only.
7 · Thrusters	Torque = force × arm. Two 1 N jets, each 1 m from the centre, firing opposite ways: **2 N·m** and no sideways push. Every puff is propellant you never get back.
8 · Why wheels burst	The rim of a 20 cm wheel at 6,000 rpm moves at 63 m/s. At 65,000 rpm it would be 680 m/s, and the stress (density × speed²) about 3.6 GPa: more than steel can take.

SEC 6 · LEC 3 · COMPARE + FACT CHECK

Five actuators, one table

The comparison to know, then where the instructor was loose, the caption errors, and the say-it-out-loud script.

📋 The actuators side by side

Actuator	How it twists	Kind	Typical torque	Uses up	The catch
Reaction wheel	changes its speed, either way	inside: swaps	0.01–0.5 N·m	electricity	saturates; bearings wear out
Momentum wheel	spins fast always; small speed changes	inside: swaps	0.01–0.5 N·m	electricity	controls one axis; saturates
Control moment gyro	tilts a constant-speed wheel	inside: swaps	100s of N·m	electricity	heavy, complex; saturates too
Magnetic torquer	electromagnet in the planet's field	outside: removes	about 0.0003 N·m	electricity	low orbit only; nothing about the field line
Thrusters	small jets off-centre, in pairs	outside: removes	0.1 to 100s of N·m	propellant	runs out; coarse; exhaust can coat optics

▲ Where the instructor was loose

- **Not said: the inside / outside split.** Wheels and CMGs only trade spin with the spacecraft; the total never changes. Only thrusters and magnetic torquers can change the total. That is why every wheel system also carries one of those. It is the most important idea in this lecture.
- **Saturation "occurs commonly due to instabilities".** It comes from steady outside nudges: sunlight pressure, thin air, the pull of gravity on a long body. They push the same way for days, and the wheel has to keep absorbing them.
- **Momentum wheel versus reaction wheel.** He blurs them. A momentum wheel keeps a large spin all the time. A reaction wheel rests near zero and goes either way.
- **"Speeds from 3,500 to 65,000 rpm."** Flight wheels run at a few thousand rpm. As far as I know, 65,000 belongs to experimental energy-storage flywheels.
- **"CMGs are a complete set of momentum wheels in all directions."** One CMG is one wheel on a gimbal. Several are needed; the Station has four.
- **"High slew rate… very time-responsive, not much lag."** Slew rate is how fast the spacecraft can turn. CMGs give big torque, so heavy craft turn quickly.
- **Magnetic torquers "align the sensor" and "work similarly to CMGs".** They are actuators, and nothing like a CMG: they push on the Earth's field, which is outside the spacecraft.
- **Cold gas "usually carbon dioxide".** Usually nitrogen. Hot-gas ones burn or decompose hydrazine.
- **Slide typos:** "Reactor wheels" = reaction wheels. "3 axis determination" = 3-axis *control*.
- **He was right** about the helicopter: the tail rotor exists to cancel the body's reaction to the main rotor.

🔎 Caption errors

Captions said	Instructor meant
"attrited determination"	**attitude** determination
"momentum Boyle's", "moments of will"	**momentum wheels**
"magnetic talks / talkers", "restoring talk"	magnetic **torquers**, restoring **torque**
"thrust us"	**thrusters**
"helicopters need to tail"	need **a tail rotor**
"the amount of the wind can blow"	the **mass of the wheel can be low**
"separation limit"	**saturation** limit
"Maybrick"	**may break**
"reactor Wilt's"	**reaction wheels**
"Control Moment Hiros", "momand Jews"	**control moment gyros**
"solar noise"	**solenoids** (coils)
"calls of hot thrusters"	**cold or hot gas** thrusters
"froggeries ontology axis"	probably the **horizontal** axis (angle θ on the slide)

🗣 Say it out loud

Friend: "How does a spacecraft turn with nothing to push against?"
You: Mostly by spinning a wheel inside. Spin the wheel one way and the spacecraft turns the other, like swinging your arms on a swivel chair. No fuel. The catch is that outside nudges make the wheels spin faster and faster until they are full. Then it has to push on something outside to empty them: a little jet, or an electromagnet leaning on the Earth's magnetic field. Big things like the Space Station use a spinning wheel they tilt, which twists much harder.

SEC 6 · LEC 3 · VOCABULARY

Every word used, in plain English

37 entries. "Repeat" = you have met it before.

Actuator — a device that pushes or twists the spacecraft. *Repeat.*

Torque — a twist: force × lever arm. Captions write 'talk'. *Repeat.*

Angular momentum — amount of spin: moment of inertia × spin rate. *Repeat.*

Conservation of angular momentum — the total spin of a closed system never changes. *Swing your arms on a swivel chair.*

Momentum wheel — a wheel kept spinning fast all the time; small speed changes turn the craft.

Reaction wheel — a wheel that rests near zero and is spun either way to turn the craft.

Flywheel — a heavy wheel built to store spin.

rpm — revolutions per minute.

DC motor — an electric motor run from steady (direct) current. It drives the wheel.

Saturation / saturation limit — the wheel has reached top speed and can absorb no more. *A full bucket.*

Momentum dumping — emptying a wheel by pushing on something outside. *Repeat.*

Instability — a tendency to drift or tip away.

Centrifugal force — the outward pull felt by anything going round. It tries to tear a fast wheel apart.

Stress — force inside a material per unit of area.

Three-axis control — controlling roll, pitch and yaw separately.

Control moment gyro (CMG) — a constant-speed wheel whose axis is tilted to make a large torque. *A tilted bicycle wheel.*

Gimbal — the pivoting mount that tilts the wheel. *Repeat.*

Slew / slew rate — a turn to a new target / how fast the craft can turn.

Tracking — keeping pointed at a moving target.

Lag — delay between command and response.

International Space Station (ISS) — the crewed station; it steers with four CMGs. *Repeat.*

Magnetic torquer (magnetorquer, torque rod) — a coil that becomes a magnet when current flows; the planet's field twists it. *A compass needle.*

Solenoid — a coil of wire.

Magnetic dipole (moment) — how strong a magnet the coil is.

Restoring torque — a twist that pulls something back into line.

Electrical current — the flow of electric charge, in amperes.

Static (no moving parts) — nothing in it moves, so nothing wears out.

Reliable — unlikely to fail.

Thruster — a small rocket engine used for pointing or small manoeuvres. *Repeat, Section 4.*

Cold-gas thruster — a valve that lets stored gas (usually nitrogen) squirt out.

Hot-gas thruster — one that heats or burns its propellant, usually hydrazine.

Moment arm — the distance from the centre of mass to where a force acts.

Couple — two equal, opposite forces on opposite sides: a pure twist.

Reaction control system (RCS) — the full set of small attitude thrusters.

Internal / external torque — a twist from inside (wheels) / from pushing on something outside.

Disturbance torque — small unwanted outside twists that never stop. *Repeat, Section 2.*

Kepler — NASA's planet-hunting telescope; it lost two of its four reaction wheels. *Who's Who.*